

FRANKLIN COUNTY AP, VT, USA

WMO: 720494

Lat: 44.933N

Lon: 73.100W

Elev: 230

StdP: 14.57

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 00152

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.8	(c) -3.9	(d) -17.6	(e) 2.1	(f) -8.2	(g) -12.3	(h) 2.8	(i) -2.8	(j) 19.5	(k) 36.9	(l) 17.5	(m) 35.6	(n) 3.1	(o) 130	(p) 0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	88.2	72.3	85.1	70.5	81.7	68.6	74.9	83.5	73.0	80.9	71.4	78.8	5.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.3	120.7	79.6	70.4	112.9	77.3	68.8	106.7	75.9	38.4	83.7	36.7	81.0	35.3	79.0	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	13.5	11.8	-18.2	92.2	5.1	3.4	-21.9	94.7	-24.8	96.7	-27.7	98.6	-31.4	101.1
			-18.0	77.6	5.0	2.6	-21.6	79.4	-24.6	80.9	-27.4	82.4	-31.0	84.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.2	18.4	21.9	30.7	44.0	58.1	65.0	71.1	69.1	61.6	49.2	37.2	25.9
	DBStd	20.43	12.78	11.42	10.87	9.00	8.42	6.55	5.89	5.22	7.86	8.51	9.45	11.63
	HDD50	3885	980	786	602	225	22	0	0	0	5	121	395	749
	HDD65	7392	1445	1206	1062	632	244	77	13	20	155	491	834	1213
	CDD50	2486	0	0	5	45	275	451	654	593	354	97	11	1
	CDD65	517	0	0	0	2	32	77	202	147	54	3	0	0
	CDH74	3999	0	0	0	27	316	619	1662	973	393	9	0	0
	CDH80	1042	0	0	0	3	79	154	521	207	78	0	0	0
Wind	WSAvg	4.6	4.9	4.9	5.4	5.7	5.1	4.1	3.5	3.5	3.9	4.5	5.1	4.4
Precipitation	PrecAvg	37.6	2.2	1.9	2.3	3.0	3.2	4.0	4.3	3.9	3.4	4.0	3.0	2.6
	PrecMax	43.7	5.7	3.1	4.5	5.2	6.9	7.6	6.7	7.4	7.6	7.4	5.2	6.0
	PrecMin	28.2	0.9	0.9	0.7	0.5	1.2	1.5	2.0	1.2	1.7	0.9	1.0	0.9
	PrecStd	4.5	1.0	0.7	0.9	1.3	1.4	1.7	1.3	1.4	1.3	1.9	1.0	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.0	54.8	69.7	77.4	87.6	89.5	93.2	88.7	88.1	75.1	64.5	55.4
		MCWB	50.5	49.5	57.7	58.7	70.6	72.4	76.2	73.0	73.5	64.8	56.4	50.5
	2%	DB	45.6	45.9	55.5	70.9	81.5	84.4	89.6	85.8	81.7	70.4	59.5	49.7
		MCWB	42.3	40.3	46.9	55.2	64.1	68.1	73.4	71.4	70.2	61.6	51.2	45.9
	5%	DB	39.7	41.0	51.6	64.4	77.1	80.6	86.2	81.8	78.9	66.3	55.2	45.8
		MCWB	36.6	36.4	44.0	52.1	61.8	66.4	71.9	68.7	68.6	59.0	49.3	43.0
	10%	DB	36.1	37.3	46.0	59.4	73.2	77.1	82.0	79.4	75.1	62.2	52.0	41.0
		MCWB	33.2	33.8	40.5	49.2	60.8	64.6	69.8	67.8	66.5	54.9	46.8	38.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.8	50.4	58.3	62.2	71.6	74.5	79.1	75.3	75.1	67.5	59.1	51.1
		MCDB	53.0	53.7	68.7	74.5	83.5	84.7	90.2	84.6	84.3	71.7	63.1	54.5
	2%	WB	42.6	40.7	49.2	58.1	68.2	71.7	76.1	73.4	72.5	64.0	53.3	46.5
		MCDB	45.3	44.8	55.0	68.0	77.4	79.8	84.4	80.6	79.5	69.0	57.5	48.9
	5%	WB	37.0	37.2	44.4	53.8	65.8	69.5	73.7	71.7	70.2	59.6	50.6	42.9
		MCDB	39.9	40.9	49.5	61.9	73.2	76.2	81.7	78.4	76.8	64.6	55.4	45.5
	10%	WB	33.1	34.3	40.9	50.5	62.6	67.3	72.0	70.2	67.9	56.4	46.5	38.1
		MCDB	35.3	37.2	46.2	58.8	70.3	74.3	79.8	76.8	73.9	60.9	51.0	40.8
Mean Daily Temperature Range	5% DB	MDBR	16.3	15.9	16.1	18.8	21.2	19.3	19.6	18.9	19.5	15.7	13.9	12.9
		MCDBR	17.0	18.7	22.6	27.7	26.1	24.0	23.0	21.4	22.5	20.4	18.9	15.4
		MCWBR	15.8	15.2	15.3	16.2	13.1	11.7	10.4	10.4	12.4	12.9	14.4	14.9
	5% WB	MCDBR	16.9	18.4	20.0	25.1	21.1	18.4	19.3	17.7	20.5	17.6	17.5	15.4
		MCWBR	16.0	15.4	15.1	16.0	12.8	10.7	10.0	9.7	12.2	13.0	14.8	15.4
Clear-Sky Solar Irradiance	taub	0.275	0.290	0.328	0.357	0.381	0.433	0.440	0.417	0.374	0.335	0.311	0.287	
	taud	2.361	2.317	2.356	2.384	2.369	2.228	2.216	2.272	2.383	2.497	2.488	2.381	
	Ebn at Noon	268	284	287	287	282	266	263	265	268	266	254	250	
	Edh at Noon	24	30	34	36	38	43	43	40	33	25	22	22	
All-Sky Solar Radiation	RadAvg	454	736	1107	1367	1602	1746	1770	1581	1267	736	456	325	
	RadStd	52	75	82	165	165	145	133	74	101	73	57	28	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+115	N/A

Nomenclature: See separate page